

MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY
COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY



DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION ENGINEERING

ET 8307: PRINCIPLE OF COMMUNICATION SYSTEM

TIME: 60 Minutes

DATE: 17th Jan, 2025 PM

INSTRUCTIONS

1. Answer **ALL** questions and **Do NOT** write anything in this question paper.
2. All answers should be **WRITTEN** in the answer sheet provided (page 4, 5 and 6)

Question 1:

Fill in the blank(s) by writing missing word (s)

[10 marks]

- i. Thermal noise is generated by the random motion of electrons in resistors and semiconductors.
- ii. The process of Modulation involves varying a carrier signal's properties based on the message signal.
- iii. In Full duplex communication, both transmission and reception occur simultaneously.
- iv. The smallest value of T_0 that satisfies $x(t) = x(t + T_0)$ for a periodic signal is called the Period.
- v. Digital communication systems are more robust against noise compared to analog systems.
- vi. A noise is an unwanted signal that distorts or interferes with the desired signal during transmission.
- vii. Source encoding is the process of converting a message signal into bits for efficient channel transmission.
- viii. Amplitude modulation involves varying the amplitude of the carrier signal based on the message signal.
- ix. The Bandwidth of a communication channel determines how well it can transmit signals without distortion.

- x. A _____ is a combination of components designed for the purpose of transmitting information from one point to another.

Question 2:

Match items in list 1 by corresponding items in list 2

[04 marks]

LIST 1	LIST 2
I. Converts analog signals to digital signals. C	A. Digital Signal
II. Introduces distortion and noise. E	B. Amplitude Modulation
III. Recover the original message. J	C. Fuse
IV. Transmits data in one direction only. N	D. Filter
V. Varies the frequency of the carrier signal. M	E. Channel
VI. Varies the amplitude of the carrier signal. B	F. Source Encoder
VII. Measure of signal degradation in a system. P	G. Transmitter
VIII. A sequence of discrete values representing information A	H. Amplifier
	I. Circuit Breaker
	J. Receiver
	K. Full Duplex Communication
	L. Data Communication
	M. Frequency Modulation
	N. Simplex Communication
	O. Digital to Analog Converter

Question 3 (a)

Write correct for the **True** statement and/or wrong for the **False** statement. [03 marks]

- In a communication system, the transmitter introduces noise to the signal. *wrong*
- Frequency modulation is achieved by varying the amplitude of the carrier wave. *wrong*
- Noise affects the performance of a communication system by distorting the transmitted signal. *correct*
- Digital communication systems are less robust against noise than analog systems. *wrong*
- A periodic signal repeats itself over a regular interval of time. *correct*
- The receiver in a communication system converts the message back to its original form. *correct*

Question 3 (b)

[03 marks]

- Describe the major types of communication modes used in modern communication systems and their applications.
- A communication channel has a signal power of and noise power of -20dBW and 2mW respectively. Calculate the signal-to-noise ratio (SNR) in decibels (dB). *7dB*
- A receiver has an input signal-to-noise ratio (S/N) of 30 and an output signal-to-noise ratio (S/N) of 20. Calculate the noise factor (F) and noise figure (in dB).

$$\text{Noise factor} = 1.5$$

$$\text{Noise figure} = 1.76$$